

# Flood plain mapping and risk assessment of Part of Krishna River using HEC-RAS and AHP Modelling.



# **Introductiopn of Resurch Paper:-**

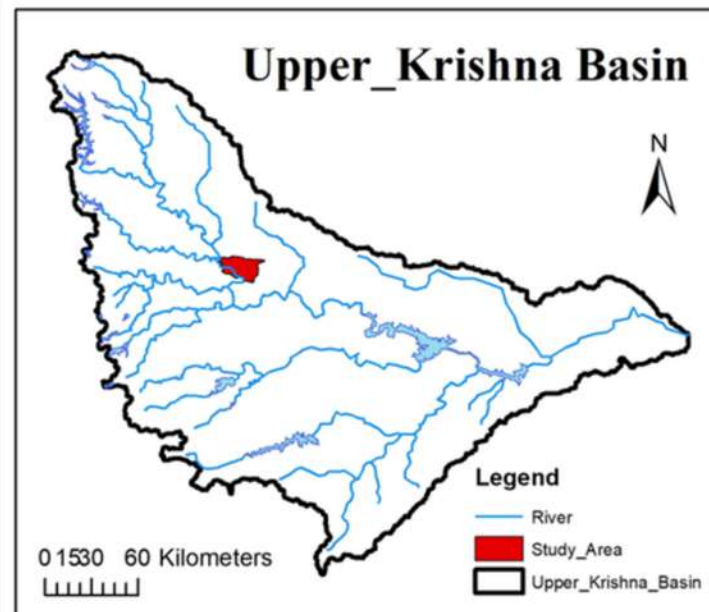
A flood is an unusually high stage in a river, normally the level at which the river overflows its banks and inundates the adjoining area. Aims to identify areas prone to flooding within a river basin by creating detailed maps of the flood plain, allowing for the evaluation of potential flood hazards and associated risks to people, property, and infrastructure, ultimately informing mitigation strategies and land use planning decisions to minimize flood impacts within vulnerable areas. This project aims to conduct a comprehensive flood plain mapping and risk assessment for the [Krishna River Basin] to accurately identify are as prone to flooding and evaluate the associated risks to communities and infrastructure. By utilizing advanced GIS technology, hydrological modeling, and detailed data analysis, the study will produce detailed flood risk maps, enabling informed land use planning and mitigation strategies to minimize the potential impacts of future flood events within vulnerable areas.

## **Objective :-**

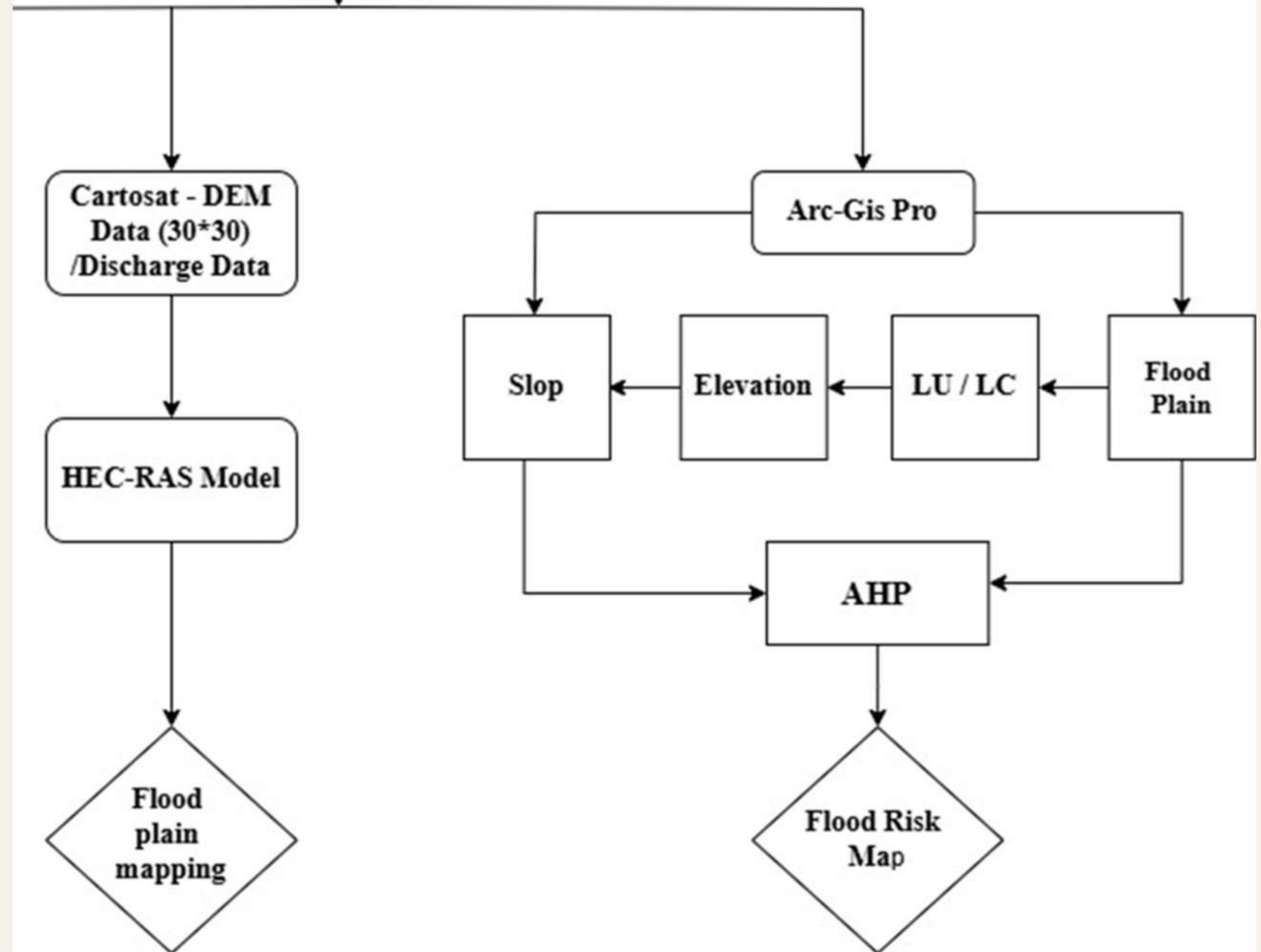
- To carried out Flood Analysis , A part of Krishna river.
- To create flood risk map using HEC-RAS and Analytical Hierarchy Process (AHP) Modelling.



# STUDY AREA



## Methodology



## SOURCE OF DATASET

| Sr. No. | Dataset Type                                     | Formats                       | Source                                                                            |
|---------|--------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------|
| 1.      | Watershed Basin                                  | Shapefile (.shp)              | <a href="https://www.hydrosheds.org">https://www.hydrosheds.org</a>               |
| 2.      | India Subdistricts Boundaries                    | Shapefile (.shp)              | <a href="https://www.surveyofindia.gov.in/">https://www.surveyofindia.gov.in/</a> |
| 3.      | Digital Elevation Model (DEM) of 30 m Resolution | Tag Image File Formant (TIFF) | <a href="https://www.asf.gov/">https://www.asf.gov/</a>                           |
| 4.      | Projection File                                  | Projection (.prj)             | <a href="https://spatialreference.org">Spatialreference.org</a>                   |
| 5.      | Discharge Data                                   | ---                           | <a href="https://wrd.maharashtra.gov.in/">https://wrd.maharashtra.gov.in/</a>     |



# STUDY AREA MAP

## KRISHNA RIVER ANKALI TO MHAISAL

